

Chapter 3 Homework Answer Sheet

Part A Multiple choice (10 points)

1-5 DEACD

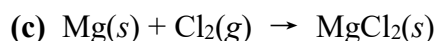
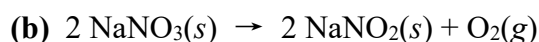
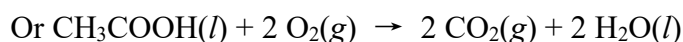
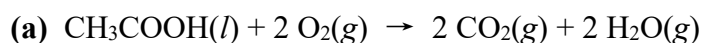
6-10 ADECB

Part B Short questions (15 points)

11. (2 points)

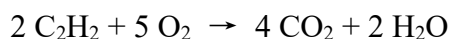
1 mol of lithium carbonate $< 9 \times 10^{23}$ molecules of carbon dioxide $< 24 \text{ g } ^{12}\text{C} < 1 \text{ mol C}_3\text{H}_8$

12. (6 points)



13. (7 points)

(a) (2 points)



$$10.0 \text{ g C}_2\text{H}_2 \times \frac{1 \text{ mol C}_2\text{H}_2}{26.04 \text{ g C}_2\text{H}_2} \times \frac{5 \text{ mol O}_2}{2 \text{ mol C}_2\text{H}_2} \times \frac{32.00 \text{ g O}_2}{1 \text{ mol O}_2} = 30.7 \text{ g O}_2 \text{ required}$$

Only 10.0 g O_2 are available, so O_2 limits.

(b) (5 points)

Because O_2 limits, 0.0 g O_2 remain.

Next, calculate the g C_2H_2 consumed and the amounts of CO_2 and H_2O produced by reaction of 10.0 g O_2 .

$$10.0 \text{ g O}_2 \times \frac{1 \text{ mol O}_2}{32.00 \text{ g O}_2} \times \frac{2 \text{ mol C}_2\text{H}_2}{5 \text{ mol O}_2} \times \frac{26.04 \text{ g C}_2\text{H}_2}{1 \text{ mol C}_2\text{H}_2} = 3.26 \text{ g C}_2\text{H}_2 \text{ consumed}$$

$$10.0 \text{ g C}_2\text{H}_2 \text{ initial} - 3.26 \text{ g consumed} = 6.74 = 6.7 \text{ g C}_2\text{H}_2 \text{ remain}$$

$$10.0 \text{ g O}_2 \times \frac{1 \text{ mol O}_2}{32.00 \text{ g O}_2} \times \frac{4 \text{ mol CO}_2}{5 \text{ mol O}_2} \times \frac{44.01 \text{ g CO}_2}{1 \text{ mol CO}_2} = 11.0 \text{ g CO}_2 \text{ produced}$$

$$10.0 \text{ g O}_2 \times \frac{1 \text{ mol O}_2}{32.00 \text{ g O}_2} \times \frac{2 \text{ mol H}_2\text{O}}{5 \text{ mol O}_2} \times \frac{18.02 \text{ g H}_2\text{O}}{1 \text{ mol H}_2\text{O}} = 2.25 \text{ g H}_2\text{O produced}$$